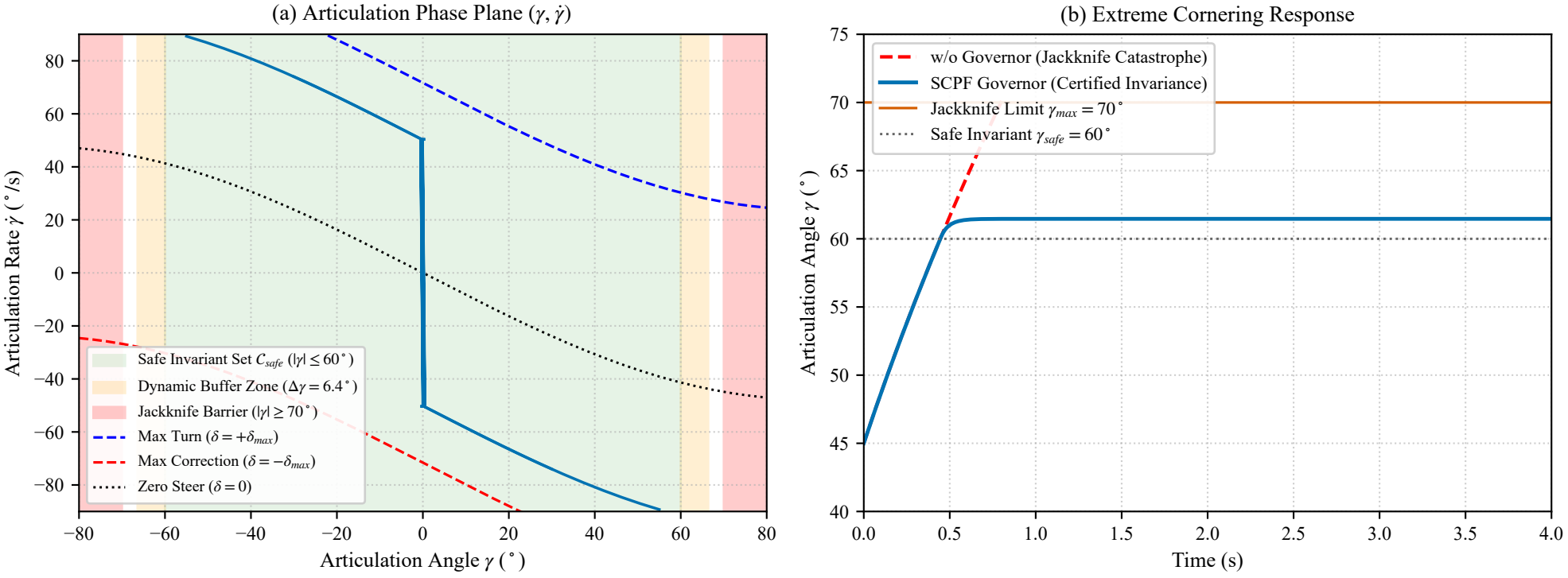

(a) Articulation Phase Plane ($\gamma, \dot{\gamma}$)
Articulation Rate $\dot{\gamma}$ (°/s)
Articulation Angle γ (°)
Safe Invariant Set C_{safe} ($|\gamma| \leq 60°$)
Dynamic Buffer Zone ($\Delta\gamma = 6.4°$)
Jackknife Barrier ($|\gamma| \geq 70°$)
Max Turn ($\delta = +\delta_{max}$)
Max Correction ($\delta = -\delta_{max}$)
Zero Steer ($\delta = 0$)
(b) Extreme Cornering Response
Articulation Angle γ (°)
Time (s)
w/o Governor (Jackknife Catastrophe)
SCPF Governor (Certified Invariance)
Jackknife Limit $\gamma_{max} = 70°$
Safe Invariant $\gamma_{safe} = 60°$